

Ruoya Li

☎ 669-301-8206 | @ ruoyali1020@gmail.com | 🔗 LinkedIn/ruoya-li | 🐙 GitHub/ruoyali7 | 📍 Sunnyvale, CA | U.S. Permanent Resident

SUMMARY

Software Engineer combining **generative AI, software engineering, and 3D creative technology**, with a background in architecture, **3D visualization, animation, and computational design**. Experienced in building AI-assisted creative tools and automated workflows spanning video production, parametric design, and digital fabrication.

SKILL

AI & Software: OpenAI API, RAG, Python, JavaScript/TypeScript, Java, React.js, Next.js, Node.js, FastAPI

3D & Visualization: Maya, ZBrush, Rhino, Revit, SketchUp, V-Ray, Enscape, Lumion, Adobe Creative Suite, After Effects, 3D Modeling, Animation, VR

Computational Design & Fabrication: Grasshopper, Rhino API, C#, KUKA Robotics, CNC, 3D Printing

Platforms & Tools: AWS, Cloudflare R2, Supabase, Railway, GitHub Actions

WORK EXPERIENCE

DramaClips 2025 – Present

Independent Software Engineer

Remote

- Built and launched an **AI-assisted short-drama content platform** using Next.js, TypeScript, Supabase, Cloudflare R2, and Railway, supporting end-to-end content production and multi-platform publishing.
- Designed **asynchronous video workflows** for ingestion, AI hook generation, asset persistence, and clipping, reducing manual processing and accelerating content from source video to publish-ready assets.
- Developed **content-production and analytics tooling** for metadata import, video asset organization, attribution tracking, and performance analytics, centralizing content operations and campaign performance data.

GEICO Summer 2024, Summer 2025

Software Engineer Intern, Technology Development Program (TDP)

Remote

- Enhanced backend services for GEICO's **B2B Commercial Digital Marketplace**, extending REST APIs and shared data contracts to enable data exchange between internal insurance platforms and external carriers.
- Implemented **carrier onboarding enhancements** by updating API contracts, DTOs, and data mappings across backend services while preserving compatibility with existing carrier integrations.
- Developed **full-stack features** for LaunchPad, an internal platform supporting commercial insurance agent workflows, using React.js, .NET, and SQL Server.

Gensler Jun 2021 – Dec 2021

Technical Designer

Beijing, China

- Created **3D models, renderings, visualizations, and animations** with Maya, Rhino, SketchUp, V-Ray, and Adobe Creative Suite to communicate design concepts and support stakeholder presentations.
- Developed **Grasshopper algorithms and C# scripts using the Rhino API** to automate parametric design and visualization workflows, reducing repetitive manual work and accelerating design iteration.

Steinberg Hart Jan 2019 – Mar 2021

Architectural Designer

San Jose, CA

- Produced **3D models, architectural renderings, and VR-based immersive visualizations** for commercial and mixed-use projects, supporting concept development, design communication, and client presentations.
- Developed **Python and JavaScript automation tools** for BIM and digital design workflows, improving efficiency across commercial, residential, and education projects with budgets up to \$100M.

PROJECTS

AI Interview Copilot: RAG-based LLM System

- Built a **RAG-based AI interview coaching system** using FastAPI, OpenAI API, and ChromaDB to generate personalized interview questions and evaluate candidate answers.
- Implemented a **two-stage retrieval pipeline** using SentenceTransformers embeddings, vector search, and a CrossEncoder reranker, with structured JSON outputs and retry handling.

EDUCATION

Johns Hopkins University 2023 – 2025

M.Sc. Computer Science (GPA: 4.0)

Southern California Institute of Architecture 2016 – 2018

M.Arch in Architecture

Kent State University 2011 – 2016

B.S. Architecture